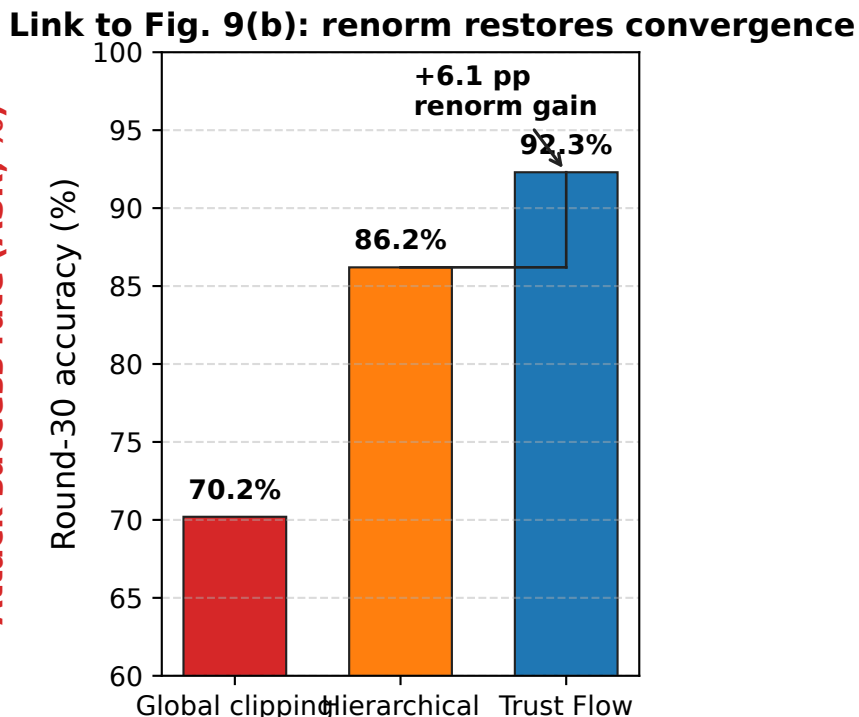
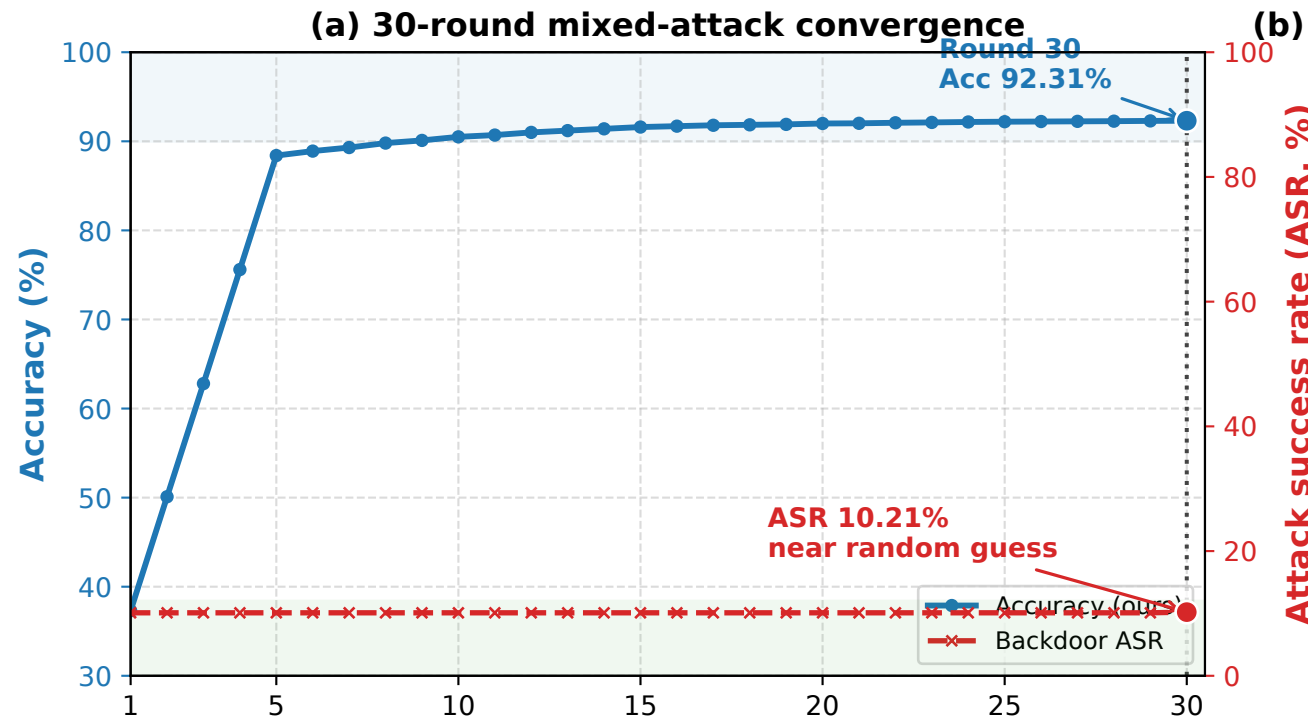


Global Robustness Under Mixed Attacks and the Role of Renormalization



Experimental setting: 20 admitted clients, 30% malicious (6/20), six mixed attack types, Dirichlet $\alpha=0.1$, 30 rounds, 5-seed mean. Key point: low ASR is not obtained by freezing learning; survivor-weight renormalization preserves optimization momentum, matching Fig. 9(b).